

Swaham Pitambar Patra

Task 1:

```
1 # Write a Python program to create a "Student" class with a private "__marks" attribute and a method to display marks.
2
3 class Student:
4     def __init__(self, marks):
5         self.__marks = marks
6
7     def display_marks(self):
8         print(f"Marks: {self.__marks}")
9
10 S1 = Student(85)
11 S1.display_marks()
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS

✓ TERMINAL

PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> & "C:\Users\Swaham Patra\AppData\Local\Programs\Python\Python314\python.exe" "c:/Users/Swaham Patra/OneDrive/ITR Python+DS/Tasks 03-07/tas
Marks: 85

Task 2:

```
1 # Create an "Employee" class with a private "__salary" attribute and methods to update and display the salary.
2
3 class Employee:
4     def __init__(self, Salary):
5         self.__Salary = Salary
6
7     def update_Salary(self, new_salary):
8         self.__Salary = new_salary
9     def display_Salary(self):
10         print(f"Salary: {self.__Salary}")
11
12 S1 = Employee(85000)
13
14 print("Before updating salary:")
15 S1.display_Salary()
16
17 print("After updating salary:")
18 S1.update_Salary(90000)
19 S1.display_Salary()
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS

✓ TERMINAL

PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS> & "C:\Users\Swaham Patra\AppData\Local\Programs\Python\Python314\python.exe" "c:/Users/Swaham Patra/OneDrive/ITR Python+DS/Ta
Before updating salary:
Salary: 85000
After updating salary:
Salary: 90000

Task 3:

```
Tasks 03-07 > task3.py > ...
1 # Write a Python program to implement encapsulation using an "ATM" class with a private "__pin" attribute.
2
3 class ATM:
4     def __init__(self, pin):
5         self.__pin = pin
6
7     def pin_check(self, entered_pin):
8         if entered_pin == self.__pin:
9             print("Welcome to the ATM!")
10        else:
11            print("PIN doesn't match. Authorization failed!! \n Try again later.")
12
13 user_pin = int(input("Set your PIN:"))
14 atm = ATM(user_pin)
15
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS

✓ TERMINAL

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python\DS> & "C:\Users\Swaham Patra\AppData\Local\Programs\Python\Python314\python.exe" "c:/Users/Swaham Patra/OneDrive/ITR Python/DS/Tasks 03-07/task3.py"
Set your PIN:1234
Confirm your PIN :1234
Welcome to the ATM!
PS C:\Users\Swaham Patra\OneDrive\ITR Python\DS> & "C:\Users\Swaham Patra\AppData\Local\Programs\Python\Python314\python.exe" "c:/Users/Swaham Patra/OneDrive/ITR Python/DS/Tasks 03-07/task3.py"
Set your PIN:1234
Confirm your PIN :2311
PIN doesn't match. Authorization failed!!
Try again later.
PS C:\Users\Swaham Patra\OneDrive\ITR Python\DS> |
```

Task 4:

```
1  # Create a "Library" class with a private "__books" attribute and methods to issue and return books.
2
3  class Library:
4      def __init__(self):
5          self.__books = ["DBMS", "MIC", "STE", "DCN"]
6
7      def issue(self, book):
8
9          if book in self.__books:
10             self.__books.remove(book)
11             print(f"Book '{book}' issued successfully.")
12             return True
13
14          else:
15             print(f"Book '{book}' is not available.")
16             return False
17
18      def return_book(self, book):
19
20          self.__books.append(book)
21          print(f"Book '{book}' returned successfully.")
22          return True
23
24  lib = Library()
25
26  while True:
27      print("\nOptions:")
28      print("1. Issue Book")
29      print("2. Return Book")
30      print("3. Exit")
31
32      choice = input("\nEnter your choice (1-3): ")
33
34      if choice == "1":
35          book = input("Enter book name to issue: ")
36          if book:
37              lib.issue(book)
38      elif choice == "2":
39          book = input("Enter book name to return: ")
40          if book:
41              lib.return_book(book)
42
43      elif choice == "3":
44          print("Thank you for using the Library System!")
45          break
46      else:
47          print("Invalid choice! Please enter 1, 2, 3, or 4.")
```

Output:

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>

Options:
1. Issue Book
2. Return Book
3. Exit

Enter your choice (1-3): 1
Enter book name to issue: MIC
Book 'MIC' issued successfully.

Options:
1. Issue Book
2. Return Book
3. Exit

Enter your choice (1-3): 2
Enter book name to return: DBMS
Book 'DBMS' returned successfully.

Options:
1. Issue Book
2. Return Book
3. Exit

Enter your choice (1-3): 3
Thank you for using the Library System!
PS C:\Users\Swaham Patra\OneDrive\ITR Python+DS>
```